MagicBox Playbook Reference

Reference for every Ansible recipe in this repository: purpose, variables, commands, tags, prerequisites, verification and operational risks. It matches the code at the time of the build; `tools/check_docs.py` fails if a role, variable or tag is missing from this document.

Pinned toolchain: ansible-core 2.21.5 (Python 3.14 `.venv` managed by uv), community.general 13.5.0, community.docker 5.4.0, community.crypto 3.5.1.

1. Overview

Item	Value
Playbooks	playbooks/site.yml (host + Portainer), playbooks/stacks.yml (default stacks)
Inventory	inventory/hosts.yml (git-ignored; template inventory/hosts.example.yml)
Group	magicbox
Project policy	inventory/group_vars/magicbox/main.yml
Secrets	inventory/group_vars/magicbox/vault.yml (ansible-vault, git-ignored)
Collections	requirements.yml, installed to .ansible/collections
Result	Portainer CE at https://magicbox.local:9443/ with environment homelab; default stacks; LAN only

Play 1, *Configure MagicBox*, runs on group `magicbox` with `become: true` and applies the roles in this order: `preflight`, `mdns`, `firewall`, `docker`, `portainer`, `portainer_init`. Play 2, *Verify Portainer from the Ansible controller*, queries Portainer from the controller over the LAN.

`playbooks/stacks.yml` has one play, *Deploy default stacks through Portainer*, that runs the `stacks` role on group `magicbox`. Run it after `site.yml`.

2. Commands

Goal	Command
Create the environment	make setup
All local checks	make check
Container converge + idempotency test	make test-container
First deployment	make deploy EXTRA="-K -e magicbox_confirm_destructive=true"
Routine deployment	make deploy EXTRA="-K"
Only one area	uv run ansible-playbook playbooks/site.yml --ask-vault-pass -K --tags docker
Dry run with diff	uv run ansible-playbook playbooks/site.yml --ask-vault-pass -K --check --diff
Verification only	make verify EXTRA="-K"
Default stacks	make stacks EXTRA="-K"
Selected stacks only	uv run ansible-playbook playbooks/stacks.yml --ask-vault-pass -K -e '{"stacks_enabled": ["postgresql"]}'
Stacks: verification only	uv run ansible-playbook playbooks/stacks.yml --ask-vault-pass -K --tags verify
Idempotency on MagicBox	make idempotency VAULT_PASSWORD_FILE=<file> EXTRA="--become-password-file=<file>"
List tasks / tags	uv run ansible-playbook playbooks/site.yml --list-tasks / --list-tags

`-K` asks for the sudo password; omit it if `<admin-user>` has passwordless sudo. A `--check` run cannot complete the `docker` and `portainer` roles on a fresh host because the packages they verify are not installed yet.

3. Tags

Tag	Selects
preflight	The preflight role
always	Also on preflight: it runs with any --tags selection, so checks and facts are always present
mdns	The mdns role
firewall	The firewall role
docker	The docker role
portainer	The portainer and portainer_init roles, including TLS tasks
portainer-init	Only the portainer_init role (site environment)
stacks	The stacks role (in playbooks/stacks.yml)
tls	Only TLS provisioning inside portainer (certificate renewal, CA fetch)
verify	Verification tasks in all roles plus the controller play; in stacks.yml, login, lookups and stack checks
controller-verify	Only the controller play; skipped by the container test

`--tags verify` runs `preflight` (tagged `always`) and every `verify` task. It changes nothing on a converged host.

4. Shared variables

Defined in `inventory/hosts.yml` (environment facts) or `inventory/group_vars/magicbox/main.yml` (project policy).

Variable	Default	Description
magicbox_lan_interface	required	LAN interface on MagicBox (ip -br link). Avahi publishes only here.
magicbox_lan_cidr	required	Private IPv4 LAN subnet allowed to reach SSH, Portainer and mDNS.
magicbox_hostname	magicbox	Hostname and mDNS name (<name>.local); single DNS label.
magicbox_expected_architecture	x86_64	Asserted CPU architecture.
magicbox_supported_os	Ubuntu 24.04/26.04, Debian 12/13	Asserted OS/version map (ADR-0002).
magicbox_confirm_destructive	false	Allows destructive steps for one run (ADR-0008). Pass with -e.
magicbox_site_name	homelab	Name of the Portainer environment (site) for the local Docker engine (ADR-0010).
magicbox_portainer_api_url	https://magicbox.local:9443	Portainer API used by portainer_init and stacks; keep in sync with portainer_https_port.
magicbox_portainer_ca_cert	/opt/magicbox/portainer/certs/ca.crt	CA used to validate the API; keep in sync with portainer_cert_dir.
portainer_admin_password	{{ vault_portainer_admin_password }}	Bootstrap admin password, from the vault.
vault_portainer_admin_password	required (vault)	At least 12 characters; in vault.yml only.
vault_postgresql_password, vault_rabbitmq_password, vault_grafana_admin_password	required for the matching stack (vault)	At least 12 characters each.

5. Role `preflight`

Purpose. Fail fast on unsupported or misconfigured hosts and prepare the baseline: OS/architecture/inventory/vault assertions, LAN CIDR validation, detection of where the SSH controller connects from, hostname, base packages.

Prerequisites. `magicbox.local` resolves from the controller (Avahi installed during the manual steps); SSH key login as a sudo-capable user; Python 3 on the host (present on Ubuntu/Debian server installs).

Variable	Default	Description
preflight_base_packages	**ca-certificates, curl, gnupg, iproute2, python3-cryptography, python3-debian, python3-requests**	**Packages needed by later roles and modules.**
preflight_manage_hostname	**true**	**Set the static hostname and 127.0.1.1 entry. false only in the test container.**

Facts set. `preflight_controller_location`: `inside`, `outside` or `unknown` (no `SSH_CLIENT`, e.g. the container test), consumed by `firewall`.

Verification. All assertions run on every play (tag `always`).

Risks. Changing `magicbox_hostname` renames the host and the mDNS name; certificates must match (`portainer_tls_dns_names` follows automatically, the existing CA's name constraints do not: rotate the CA, installation guide 8.5).

6. Role `mdns`

Purpose. Publish `magicbox.local` on the LAN with Avahi (ADR-0003). MagicBox has no fixed IP: Avahi re-announces the name whenever DHCP changes the address. Avahi is first installed by hand (installation guide 1.5) so Ansible can connect by name; this role then owns its configuration. Installs `mdns_packages`, disables systemd-resolved's mDNS/LLMNR responders when systemd-resolved is installed, templates `/etc/avahi/avahi-daemon.conf` (host name, `allow-interfaces`, no HINFO/workstation publishing, no reflector, no wide-area), and enables `avahi-daemon`.

Variable	Default	Description
mdns_use_ipv6	**false**	**Publish AAAA records. Off because the firewall only allows the IPv4 LAN.**
mdns_packages	**avahi-daemon, avahi-utils, libnss-mdns**	**Responder, CLI tools, NSS module for local resolution.**

Handlers. *Restart systemd-resolved*, *Restart avahi-daemon*.

Verification. `avahi-resolve-host-name -4 magicbox.local` must return an address (retries 5 x 2 s).

Risks. Avahi renames the host to `magicbox-2.local` if another device already claims the name; TLS then fails the name check. Restricting to `magicbox_lan_interface` means the name is invisible on other interfaces.

7. Role `firewall`

Purpose. Enable ufw with default deny inbound and allow only SSH, `firewall_lan_tcp_ports` and `firewall_lan_udp_ports` from `magicbox_lan_cidr` (ADR-0004). Rules persist across reboots.

Variable	Default	Description
firewall_ssh_port	**22**	**SSH port allowed from the LAN.**
firewall_lan_tcp_ports	**[9443]**	**TCP ports allowed from the LAN (Portainer).**
firewall_lan_udp_ports	**[5353]**	**UDP ports allowed from the LAN (mDNS).**
firewall_logging	**low**	**ufw logging level.**

Variable	Default	Description
firewall_docker_user_filter	**true**	**Manage the DOCKER-USER block in /etc/ufw/after.rules that limits Docker-published ports to the LAN (ADR-0012).**

Safety rails. Refuses first activation unless `magicbox_confirm_destructive=true`; always refuses when `preflight_controller_location` is `outside`.

Handlers. *Reload ufw* when the DOCKER-USER block changes.

Verification. `iptables -S DOCKER-USER` contains the LAN RETURN rule and the interface DROP rule. `sudo ufw status verbose` (manual); the controller play proves 9443 is reachable from the LAN.

Risks. A wrong `magicbox_lan_cidr` can lock out SSH from other subnets (console recovery: installation guide 8.2). Rules for ports no longer listed are not removed automatically; delete them with `sudo ufw delete <n>`. Ports published by containers bypass ufw's INPUT rules; the DOCKER-USER block restricts them to `magicbox_lan_cidr` instead. Setting `firewall_docker_user_filter: false` exposes them to any source that can route to MagicBox.

8. Role `docker`

Purpose. Install Docker Engine from Docker's apt repository with a verified signing key and a major-version pin, configure the daemon, and enable `containerd` and `docker` at boot (ADR-0002).

Variable	Default	Description
docker_major_version	**"29"**	**apt preference pin 5:29.* for docker-ce, cli, rootless-extras.**
docker_apt_key_url	**https://download.docker.com/linux/<distro>/gpg**	**Signing key URL.**
docker_apt_key_fingerprint	**9DC858229FC7DD38854AE2D88D81803C0EBFCD88**	**Expected key fingerprint; mismatch aborts.**
docker_apt_key_path	**/etc/apt/keyrings/docker.asc**	**Keyring path referenced by the deb822 source.**
docker_packages	**docker-ce, docker-ce-cli, containerd.io, docker-buildx-plugin, docker-compose-plugin**	**Installed packages.**
docker_conflicting_packages	**docker.io, docker-doc, docker-compose, docker-compose-v2, podman-docker, containerd, runc**	**Removed only with confirmation.**
docker_log_max_size	**10m**	**json-file log rotation size.**
docker_log_max_file	**3**	**json-file rotated files kept.**

`daemon.json` also sets the log option `tag` to the container name, which Alloy uses as the `container` label (ADR-0013).

Files. `/etc/apt/sources.list.d/docker.sources`, `/etc/apt/preferences.d/docker-ce.pref`, `/etc/docker/daemon.json` (validated with `dockerd --validate` before replacement; `live-restore` enabled).

Handlers. *Restart docker* (containers survive thanks to live-restore).

Verification. No `dockerd` TCP listener (`ss -Hltnp`).

Risks. Removing conflicting packages stops workloads that used them. A daemon restart briefly interrupts the Docker API. A major upgrade requires changing `docker_major_version` deliberately.

9. Role `portainer`

Purpose. Run Portainer CE with persistent data, HTTPS on 9443 using the MagicBox CA, plain HTTP disabled, and a pre-set admin password (ADR-0004, ADR-0005, ADR-0006). Container: `portainer`, image `portainer/portainer-ce:2.45.3`, restart policy `unless-stopped`, `network_mode: host`; mounts the Docker socket, the `portainer_data` volume, the certificate directory (read-only) and the password file (read-only).

Variable	Default	Description
portainer_image	**portainer/portainer-ce**	**Image repository.**
portainer_version	**"2.45.3"**	**Image tag (LTS); also asserted against the API.**
portainer_container_name	**portainer**	**Container name.**
portainer_https_port	**9443**	**HTTPS port (keep in firewall_lan_tcp_ports).**
portainer_data_volume	**portainer_data**	**Persistent named volume mounted at /data.**
portainer_base_dir	**/opt/magicbox/portainer**	**Host base directory.**
portainer_cert_dir	**<base>/certs**	**TLS material (0700).**
portainer_secret_dir	**<base>/secrets**	**Admin password file (0700).**
portainer_tls_mode	**local_ca**	**local_ca or provided.**
portainer_tls_dns_names	**magicbox.local, magicbox**	**Certificate SANs and CA name constraints.**
portainer_tls_ip_addresses	**[]**	**Optional IP SANs/constraints. Keep empty: the IP is dynamic (DHCP).**
portainer_tls_ca_common_name	**MagicBox Local CA (magicbox)**	**CA subject CN.**
portainer_tls_ca_valid_for	**+3650d**	**CA lifetime.**
portainer_tls_cert_valid_for	**+397d**	**Server certificate lifetime.**
portainer_tls_renew_before	**+30d**	**Re-issue window before expiry.**
portainer_tls_ca_fetch_dest	**artifacts/magicbox-local-ca.crt**	**Controller copy of the CA for clients.**
portainer_tls_provided_cert_src	**empty**	**provided mode: server certificate path on the controller.**
portainer_tls_provided_key_src	**empty**	**provided mode: server key path (keep it vaulted).**
portainer_tls_provided_ca_src	**empty**	**provided mode: issuing CA path.**

TLS tasks (`tls_local_ca.yml`). CA key (ECDSA P-384) and self-signed CA (`CA:TRUE, pathlen:0`, critical name constraints) created once; server key (ECDSA P-256), CSR and certificate (EKU `serverAuth`) signed by the CA and re-issued when within `portainer_tls_renew_before` of expiry or when the SANs change; CA fetched to the controller.

Handlers. *Restart portainer* when TLS material changes.

Verification. `GET https://magicbox.local:9443/api/system/status` from MagicBox with CA validation (retries 20 x 3 s) and an assertion that the reported version equals `portainer_version`. The controller play repeats the request over the LAN.

Risks. Portainer controls the Docker socket and is root-equivalent: every Portainer admin is root on MagicBox. Changing the vault password after first start does not change the admin password. Deleting `portainer_cert_dir` creates a new CA that clients must trust again. Deleting the `portainer_data` volume erases all Portainer settings; the playbook never does this.

10. Role `portainer_init`

Purpose. Initialise Portainer through its API: log in as admin with the vault password and ensure one local Docker environment named `magicbox_site_name` (`homelab`). It renames an existing local-socket environment, or creates one (ADR-0010). `tasks/api_login.yml` is reused by `stacks`.

Variable	Default	Description
portainer_init_api_user	**admin**	**Portainer user for API calls; password is portainer_admin_password.**

Prerequisites. The `portainer` role has run (Portainer up, CA present).

Verification. The environment exists and its URL is `unix:///var/run/docker.sock`.

Risks. Fails at login if the admin password was changed in the UI and not in the vault. Changing `magicbox_site_name` after stacks exist creates a new, empty environment; rename the old one in the UI first.

11. Role `stacks` (playbooks/stacks.yml)

Purpose. Deploy the default services as Portainer standalone stacks in the site environment (ADR-0011). For each name in `stacks_enabled` it renders `roles/stacks/templates/<name>.yml.j2`, passes the secrets as stack environment variables, and creates the stack, or updates it when the compose file or environment differs. Config files for `observability` are written to `stacks_base_dir`; their checksum (`MAGICBOX_CONFIG_REVISION` label) forces a redeploy when they change.

Stack	Containers	Published ports	Volumes
rabbitmq	rabbitmq	5672, 15672	rabbitmq_data
postgresql	postgresql	5432	postgresql_data
ollama	ollama	11434	ollama_data
observability	grafana, loki, alloy	3000 (Grafana HTTPS)	grafana_data, loki_data, alloy_data

Variable	Default	Description
stacks_enabled	rabbitmq, postgresql, ollama, observability	Stacks to manage. Removing one does not delete it.
stacks_base_dir	/opt/magicbox/stacks	Host directory for mounted config files.
stacks_rabbitmq_image	rabbitmq:4.3.7-management	RabbitMQ image.
stacks_postgresql_image	postgres:18.6	PostgreSQL image (newest major, supported to 2030).
stacks_ollama_image	ollama/ollama:0.40.12	Ollama image.
stacks_grafana_image	grafana/grafana:13.2.5	Grafana image.
stacks_loki_image	grafana/loki:3.7.9	Loki image.
stacks_alloy_image	grafana/alloy:v1.21.2	Alloy image (Promtail replacement, ADR-0013).
stacks_rabbitmq_amqp_port	5672	Host port for AMQP.
stacks_rabbitmq_management_port	15672	Host port for the RabbitMQ UI/API (HTTP).
stacks_postgresql_port	5432	Host port for PostgreSQL.
stacks_ollama_port	11434	Host port for the Ollama API.
stacks_grafana_port	3000	Host port for Grafana (HTTPS).
stacks_rabbitmq_user	magicbox	RabbitMQ default user.
stacks_rabbitmq_password	{{ vault_rabbitmq_password }}	RabbitMQ password (first start only).
stacks_postgresql_user	magicbox	PostgreSQL superuser.
stacks_postgresql_database	magicbox	Database created at first start.
stacks_postgresql_password	{{ vault_postgresql_password }}	PostgreSQL password (first start only).
stacks_grafana_admin_password	{{ vault_grafana_admin_password }}	Grafana admin password (first start only).
stacks_ollama_models	[]	Models to pull after deployment, e.g. ["llama3.2:3b"]; never deleted.
stacks_loki_retention	168h	Loki log retention.

Internal data (`roles/stacks/vars/main.yml`, not meant to be overridden): `stacks_env` (environment per stack), `stacks_required_secrets`, `stacks_containers`.

Prerequisites. `playbooks/site.yml` completed (Portainer, `homelab` environment, DOCKER-USER filter); the vault contains the stack passwords.

Verification. All stack containers running, and healthy where a healthcheck exists (up to 15 minutes, to cover first image pulls). PostgreSQL `pg_isready`. RabbitMQ `/api/overview` with the vault credentials. Ollama `/api/version`. Grafana `/api/health` over HTTPS with the MagicBox CA. Loki has a `container` label value `loki`, which proves Alloy -> Loki -> Grafana works.

Risks.

- Passwords are applied only on first initialisation of each data volume. Changing a vault value later breaks login and verification until the service password is changed too.
- Stack environments (including passwords) are stored in Portainer's database.
- Edits made to these stacks in the Portainer UI are overwritten by the next run.
- RabbitMQ, PostgreSQL and Ollama use plain-text protocols on the LAN; Ollama has no authentication. Access is limited to the LAN by DOCKER-USER.
- Grafana uses a copy of the Portainer certificate. After a certificate renewal by `site.yml`, run `stacks.yml` to refresh the copy.
- First deployment downloads about 4 GB (Ollama is about 3.4 GB).

12. Testing the recipes

Check	Command	Proves
Syntax	make syntax	Playbook parses with the example inventory
Lint	make lint	yamllint + ansible-lint production profile
Secrets	make secrets-check	No plaintext vault values or private keys tracked
Docs	make docs-check	Links, ADR registry, this reference covers all roles/variables/tags
Container	make test-container	Full converge and changed=0 second run in an Ubuntu 24.04 systemd container (arm64 or amd64 per workstation)
Container + stacks	make test-container ARGS="--with-stacks"	Also runs stacks.yml twice (deploy + changed=0)
Host	make deploy, make idempotency, make verify	Real behaviour on MagicBox

The container test overrides `magicbox_expected_architecture`, `magicbox_lan_interface`, `magicbox_lan_cidr` and sets `preflight_manage_hostname: false`; it cannot prove LAN mDNS or reachability from other machines.

13. Changing the recipes

1. Edit the role; keep variables prefixed and documented in `defaults/main.yml`.
2. Update this reference (roles, variables, tags, risks) and, for significant decisions, add an ADR and list it in the registry and `CLAUDE.md`.
3. `make manuals check test-container`.
4. Commit sources and rebuilt PDFs together.